to circulate in the environment," says Dr Mathur.

## WHO IS AT RISK?

Those in the ICU face the highest risk—their hospitalisation is longer, their expenses are higher as more expensive, stronger drugs need to be used, and if they contract sepsis during their stay, their risk of mortality becomes extremely high. "Neonates or anyone with weakened immunity, such as people on chemotherapy or those with autoimmune disorders, also fall in this bracket," says Dr Yatin Mehta, chairman of the Institute of Critical Care and Anaesthesiology at Medanta Hospital, Gurugram, and president of the Sepsis Society of India. For him, the most concerning cases are those where a patient can be saved with an antibiotic, but the drug just doesn't work.

What is the solution then? Regulating antibiotic use, for one. A *Lancet* study published in 2022 found that although the per-capita private-sector consumption rate of antibiotics in India is relatively low compared to other nations, the country consumes a large volume of broad-spectrum antibiotics. "Antibiotics are a science. They are not like paracetamol for giving instant relief," says Dr Aditya S. Chowti, a senior consultant with the Internal Medicine wing of Fortis Hospital, Bengaluru.

In the US and Europe, antibiotic misuse is regulated by a system of surveillance, which has so far been missing in India. However, in a joint, state-backed effort between AIIMS, the ICMR and the Centers for Disease Control and Prevention (CDC) in the US, the newly-formed Healthcare-Associated Infection Surveillance-India (HAI-Surveillance) aims to strengthen surveillance of healthcare-associated infections (HAIs) and study AMR and HAI threats in India.

While the effort is commendable, the difficulty in treating drug-resistant bugs is compounded by the fact that the pipeline for new antibiotics is running dry (see *The Antibiotic Crunch*). Experts say pharmaceutical firms focus more on diabetes and hypertension drugs, which have a longer usage life. "Antibiotics need financial investment and time and their lifespan is limited now due to high rates of resistance. If the industry stops producing new drugs, we will be in serious trouble," says Dr Mathur.

## HOW BACTERIA FIGHT BACK

Resistant traits may be passed on from generation to generation of bacteria in the following ways

- **Transduction:** Resistant genes may get transferred via phages, that is, viruses that specifically infect bacteria and can carry some DNA with them
- **Conjugation:** When bacteria come in direct contact with one another
- **Transformation:** Genes released from nearby germs may be picked up by another living bacteria

## DEFENCE TACTICS

Bacteria employ five major types of 'resistance mechanisms'

- **Changing or destroying antibiotics** by secreting certain enzymes or proteins
- **Restricting the entry of antibiotics** into their cells by developing a protective covering
- **Pumping out antibiotics** that enter the cells
- **Changing an antibiotic's target** so that the drug can't find the bacteria
- **Altering themselves** to avoid using the cell process that the drug targets

## STAYING SAFE

- **Washing hands regularly,** consuming fully-cooked meats, having water only from trusted, contamination-free sources and timely vaccination for some bacterial infections such as diphtheria, cholera, pneumonia and tetanus may help

## WHAT'S NEXT?

Hospital ICUs are hotbeds of bacterial infections. Hence, the first step towards saving high-risk patients is reducing HAIs. The CDC trained over 150 professionals from the ICMR and the National Centre for Disease Control in HAI surveillance in 2018. Since then, over 30 hospitals have started such surveillance.

Another measure that a few hospitals are taking is finding out what bugs are entering the ICU and studying how to treat bacterial infections should they arise. "Most hospitals don't have good microbiology labs and aren't able to give evidence-based antibiotics," says Dr Balaji. In Delhi, however, two hospitals have altered their approach. Medanta, for example, tests patients for resistant bacteria before admitting them in the ICU. If a patient is positive, they are isolated from others to prevent spread. At Max, Saket, only a special committee can prescribe high-end antibiotics. "Our only weapon now is diagnosis," says Dr Omender Singh of Max Hospital. "Even with drug-resistant bacteria, testing gives a clearer picture of which alternative antibiotics to use."

India does have a National Action Plan for AMR, on the basis of which some states have developed their own strategies. But more must be done. The country already has over 120,000 cases of drug-resistant tuberculosis. There have also been reports of cholera, typhoid and salmonella infections growing resistant to certain drugs. New antibiotics are few and far between. If we don't act now, more lives will be lost needlessly. ■